



Object-Oriented Programming



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Agenda

1. Define Class
2. Attributes and Methods
3. Overload Methods
4. Static Class/Methods
5. References





Class

Define Class

```
1 package packageName;  
2  
3 public class className {  
4     // Define attributes  
5  
6     // Define methods  
7  
8 }
```

```
1 package Exercise01;  
2  
3 public class Fraction{  
4     // Define attributes  
5  
6     // Define methods  
7  
8 }
```

2

Attributes & Methods

Attributes

Define Attributes

```
1 package packageName;  
2  
3 public class className {  
4     // Define attributes  
5     private DataType attrubuteName_1;  
6     ...  
7     private DataType attrubuteName_n;  
8     // Define methods  
9     ...  
10 }
```

Define attributes

```
1 package Exercise01;  
2  
3 public class Fraction{  
4     // Define attributes  
5     private int numerator;  
6     private int denominator;  
7     // Define methods  
8     ...  
9 }
```

Define attributes

```
1 package Exercise02;  
2  
3 public class IntegerArray{  
4     // Define attribute  
5     private int [] arr;  
6     // Define methods  
7  
8 }
```

```
1 package Exercise03;  
2  
3 public class FractionArray{  
4     // Define attribute  
5     private Fraction [] arr;  
6     // Define methods  
7  
8 }
```

Constructors

Define constructors

```
1 package packageName;
2 public class className{
3     // Define attribute
4     private DataType attributeName_1; . . .
5     private DataType attributeName_N;
6     // Define default constructor
7     public className() {
8         this.attributeName_1 = defaultValue; . . .
9         this.attributeName_N = defaultValue;
10    }
11    // Define constructor with parameters
12    public className(DataType attributeName_1, ..., DataType attributeName_N) {
13        this.attributeName_1 = attributeName_1; . . .
14        this.attributeName_N = attributeName_N;
15    }
16    // Define copy constructor
17    public className(className object) {
18        this.attributeName_1 = object.attributeName_1; . . .
19        this.attributeName_N = object.attributeName_N;
20    }
21 }
```

Define constructors

```
1 package Exercises01;
2 public class Fraction{
3     // Define attributes
4     private int numerator;
5     private int denominator;
6     // Define default constructor
7     public Fraction() {
8         this.numerator = 1;
9         this.denominator = 2;
10    }
11    // Define constructors with parameters
12    public Fraction(int numerator, int denominator) {
13        this.numerator = numerator;
14        this.denominator = denominator;
15    }
16    // Define copy constructors
17    public Fraction(Fraction object) {
18        this.numerator = object.numerator;
19        this.denominator = object.denominator;
20    }
21 }
```

Define constructors

```
1 package Exercise01;
2
3 public class Main {
4
5     public static void main(String[] args) {
6         Fraction obj_1=new Fraction();
7         Fraction obj_2=new Fraction(2,3);
8         Fraction obj_3=new Fraction(obj_2);
9     }
10
11 }
```

Define constructors

```
1 package Exercise03;
2 public class FractionArray {
3     // Define attributes
4     private Fraction[] arr;
5     // Define default constructor
6     public FractionArray() {
7         this.arr = null;
8     }
9     // Define constructor with parameters
10    public FractionArray(Fraction... arr) {
11        this.arr = new Fraction[arr.length];
12        for (int i = 0; i < arr.length; i++) {
13            this.arr[i] = new Fraction(arr[i]);
14        }
15    }
16    // Define copy constructor
17    public FractionArray(FractionArray arrFraction) {
18        this.arr = new Fraction[arrFraction.arr.length];
19        for (int i = 0; i < arrFraction.arr.length; i++) {
20            this.arr[i] = new Fraction(arrFraction.arr[i]);
21        }
22    }
23 }
```

Define constructors

```
1  package Exercise03;
2  public class Main {
3      public static void main(String[] args) {
4          // use default constructor
5          FractionArray obj_1 = new FractionArray();
6          // use constructor with parameters
7          FractionArray obj_2 = new FractionArray(new Fraction(2,3));
8          FractionArray obj_3 = new FractionArray(new Fraction(1,3),
9                                                  new Fraction(4,3),
10                                                 new Fraction(5,9));
11
12          Fraction[] arr = new Fraction[2];
13          arr[0] = new Fraction(2,4);
14          arr[1] = new Fraction(3,7);
15          FractionArray obj_4 = new FractionArray(arr);
16          // use copy constructor
17          FractionArray obj_5=new FractionArray(obj_3);
18      }
```

Define constructors

```
1 package Exercise02;
2 public class IntegerArray {
3     // Define attributes
4     private int[] arr;
5     // Define default constructor
6     public IntegerArray() {
7         this.arr = null;
8     }
9     // Define constructor with parameters
10    public IntegerArray(int... arr) {
11        this.arr = new int[arr.length];
12        for (int i = 0; i < arr.length; i++) {
13            this.arr[i] = arr[i];
14        }
15    }
16    // Define copy constructor
17    public IntegerArray(IntegerArray object) {
18        this.arr = new int[object.arr.length];
19        for (int i = 0; i < object.arr.length; i++) {
20            this.arr[i] = object.arr[i];
21        }
22    }
23 }
```

Define constructors

```
1 package Exercise02;
2 public class Main {
3     public static void main(String[] args) {
4         // use default constructor
5         IntegerArray obj_1=new IntegerArray();
6         // use constructor with parameters
7         IntegerArray obj_2=new IntegerArray(3);
8         IntegerArray obj_3=new IntegerArray(3, 2, 8);
9         IntegerArray obj_4=new IntegerArray(4, 9, 1, 5);
10        int[] arr = {3,4,7};
11        IntegerArray obj_5=new IntegerArray(arr);
12        // use copy constructor
13        IntegerArray obj_6=new IntegerArray(obj_3);
14    }
15 }
```

Getter/Setter

Define get/set

```
1 package packageName;
2 public class className{
3     // Define attributes
4     private DataType attribute_1; . . .
5     private DataType attribute_N;
6     // Define constructors
7     ...
8     // Define get/set
9     public DataType getAttribute_1() {
10         return this.attribute_1;
11     }
12     public void setAttribute_1(DataType attribute_1) {
13         this.attribute_1 = attribute_1;
14     } . . .
15     public DataType getAttribute_N() {
16         return this.attribute_N;
17     }
18     public void setAttribute_N(DataType attribute_N) {
19         this.attribute_N = attribute_N;
20     }
21 }
```

Define get/set

```
1 public class Fraction{
2     // Define attributes
3     private int numerator;
4     private int denominator;
5     // Define constructors
6     . . .
7     // Define get/set
8     public int getNumerator() {
9         return this.numerator;
10    }
11    public void setNumerator(int numerator) {
12        this.numerator = numerator;
13    }
14    public int getDenominator() {
15        return this.denominator;
16    }
17    public void setDenominator(int denominator) {
18        if (denominator != 0) {
19            this.denominator = denominator;
20        }
21    }
22 }
```

Define get/set

```
1 package Exercise01;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Fraction obj_1 = new Fraction();
8         System.out.println(obj_1.getNumerator() + "/" + obj_1.getDenominator());
9         obj_1.setNumerator(3);
10        obj_1.setDenominator(4);
11        System.out.println(obj_1.getNumerator() + "/" + obj_1.getDenominator());
12
13    }
14 }
15
16
```

Define get/set

```
1 public class FractionArray {
2     // Define attributes
3     private Fraction[] arr;
4     // Define constructors
5     . . .
6     // Define get/set
7     public Fraction[] getArr() {
8         return this.arr;
9     }
10    public void setArr(Fraction... arr) {
11        this.arr = new Fraction[arr.length];
12        for (int i = 0; i < arr.length; i++) {
13            this.arr[i] = new Fraction(arr[i]);
14        }
15    }
16    . . .
17 }
```

Define get/set

```
1      . . .
2      public void set(int index, Fraction value) {
3          if (0 <= index && index < this.arr.length) {
4              //this.arr[index] = new Fraction(value);
5              this.arr[index]=value;
6          }
7      }
8
9      public Fraction get(int index) {
10         Fraction obj = null;
11         if (0 <= index && index < this.arr.length) {
12             //obj = new Fraction(this.arr[index]);
13             ps = this.arr[index];
14         }
15         return obj;
16     }
17 }
```

Define get/set

```
1 public static void main(String[] args) {
2     FractionArray obj = new FractionArray(
3         new Fraction(1, 3),
4         new Fraction(4, 3),
5         new Fraction(5, 9));
6     // use GET
7     Fraction[] arr = obj.getArr();
8     int index = 1;
9     Fraction f_obj = obj.get(index);
10    // user SET
11    Fraction[] arrFraction = new Fraction[2];
12    arrFraction[0] = new Fraction(2, 4);
13    arrFraction[1] = new Fraction(3, 7);
14    obj.setArr(arrFraction);
15    index=0;
16    obj.set(index, new Fraction(8, 5));
17 }
```

Define get/set

```
1 package Exercise02;
2 public class IntegerAraay {
3     // Define attributes
4     private int[] arr;
5     // Define constructor
6     . . .
7     // Define get/set
8     public int[] getArr() {
9         return arr;
10    }
11    public void setArr(int ... arr) {
12        this.arr = new int[arr.length];
13        for (int i = 0; i < arr.length; i++) {
14            this.arr[i] = rr[i];
15        }
16    }
17    . . .
18 }
```

Define get/set

```
1      . . .
2      public void set(int index, int value) {
3          if (0 <= index && index < this.arr.length) {
4              this.arr[index]=value;
5          }
6      }
7
8      public Integer get(int index) {
9          Integer value = null;
10         if (0 <= index && index < this.arr.length) {
11             value = this.arr[index];
12         }
13         return value;
14     }
15 }
```

Define get/set

```
1 public static void main(String[] args) {
2     IntegerArray obj = new IntegerArray( 3, 9, 5);
3     // use GET
4     int[] arr_1 = obj.getArr();
5     int index = 1;
6     Integer value = obj.get(index);
7     // use SET
8     int[] arr_2 = {9,6,8};
9     obj.setArr(arr_2);
10    index = 0;
11    obj.set(index, 7);
12 }
```

Input/Out Console

Define input/out console

```
1 public class Fraction{
2     // Define attributes
3     private int numerator;
4     private int denominator;
5     . . .
6     public void input(String message) {
7         System.out.println(message);
8         Scanner scan = new Scanner(System.in);
9         System.out.print("Numerator: ");
10        this.tuSo = Integer.parseInt(scan.nextLine());
11        System.out.print("Denominator: ");
12        this.mauSo = Integer.parseInt(scan.nextLine());
13    }
14
15    public void output(String message){
16        System.out.println(message);
17        String str = String.format("%d/%d", this.numerator, this.denominator);
18        System.out.println(str);
19    }
20 }
21 }
```

Define input/out console

```
1 package Exercise01;
2
3 public class Main {
4
5     public static void main(String[] args) {
6
7         Fraction obj_1=new Fraction();
8         obj_1.input("Input Fraction 1");
9         obj_1.output("Output Fraction 1");
10
11        Fraction obj_2=new Fraction();
12        obj_2.input("Input Fraction 2");
13        obj_2.output("Output Fraction 2");
14
15    }
16 }
```

Define input/out console

```
1 package Exercise03;
2 public class FractionArray {
3     // Define attributes
4     private Fraction[] arr;
5     . . .
6     public void input(String message) {
7         Scanner scan = new Scanner(System.in);
8         System.out.println(message);
9         System.out.print("Enter number of items: ");
10        int n = Integer.parseInt(scan.nextLine());
11        this.arr = new Fraction[n];
12        for(int i = 0; i < this.arr.length; i++){
13            this.arr[i] = new Fraction();
14            this.arr[i].input("Fraction "+i);
15        }
16    }
17    . . .
18 }
```

Define input/out console

```
1      . . .
2      public void output(String message) {
3          System.out.println(message);
4          int n = this.arr.length;
5          System.out.println("Number of items: " + n);
6          for(int i = 0; i < this.arr.length; i++){
7              this.arr[i].xuất("Fraction " + i);
8          }
9      }
10 }
```

Define input/out console

```
1 public static void main(String[] args) {  
2  
3     FractionArray obj = new FractionArray();  
4     obj.input("Input Fraction array");  
5     obj.output("Output Fraction array");  
6  
7 }
```

Define input/out console

```
1 package Exercise02;
2 public class IntegerArray {
3     // Define attributes
4     private int [] arr;
5     . . .
6     public void input(String message) {
7         Scanner scan = new Scanner(System.in);
8         System.out.println(message);
9         System.out.print("Enter number of items: ");
10        int n = Integer.parseInt(scan.nextLine());
11        this.arr = new int[n];
12        for(int i = 0; i < this.arr.length; i++) {
13            System.out.print("item " + i + ": ");
14            this.arr[i]=Integer.parseInt(scan.nextLine())
15        }
16    }
17    . . .
}
```

Define input/out console

```
1      . . .
2      public void output(String message) {
3          System.out.println(message);
4          int n = this.arr.length;
5          System.out.println("Number of items: " + n);
6          for(int i = 0; i < this.arr.length; i++) {
7              System.out.print("Item " + i + ": ");
8              System.out.println(this.arr[i]);
9          }
10     }
11 }
```

Define input/out console

```
1 public static void main(String[] args) {  
2  
3     IntegerArray obj = new IntegerArray();  
4     msn.input("Input integer array");  
5     msn.output("Output integer array");  
6  
7 }
```

Business methods

Define Business methods

```
1 public class Fraction{
2     // Define attributes
3     private int numerator;
4     private int denominator;
5     . . .
6     public Fraction sum(Fraction obj){
7         Fraction result = new Fraction();
8         result.numerator = this.numerator * obj.denominator + this.denominator *
9 obj.numerator;
10        result.denominator = this.denominator * obj.denominator;
11        return result;
12    }
13    . . .
}
```

Define Business methods

```
1 package Exercise01;
2 public class Main {
3     public static void main(String[] args) {
4
5         Fraction obj_1 = new Fraction();
6         obj_1.input("Input Fraction 1");
7         obj_1.output("Output Fraction 1");
8
9         Fraction obj_2 = new Fraction();
10        obj_2.input("Input Fraction 2");
11        obj_2.output("Output Fraction 2");
12
13        Fraction result = obj_1.sum(obj_2);
14        result.output("Sum of two fractions");
15    }
16 }
```

Define Business methods

```
1 package Exercise03;
2 public class FractionArray {
3     // Define attributes
4     private Fraction[] arr;
5     . . .
6     public Fraction sum() {
7         Fraction s = this.arr[0];
8         for(int i = 1; i < this.arr.length; i++) {
9             s=s.sum(this.arr[i]);
10        }
11    }
12    . . .
13 }
```

Define Business methods

```
1 public static void main(String[] args) {  
2  
3     FractionArray obj = new FractionArray();  
4     obj.input("Input Fraction array");  
5     obj.output("Output Fraction array");  
6     Fraction s = obj.sum();  
7     s.output("Sum of all items");  
8  
9 }
```

Define Business methods

```
1 package Exercise02;
2 public class IntegerArray {
3     // Define attributes
4     private int[] arr;
5     . . .
6     public int sum() {
7         int s = 0;
8         for(int i = 0; i < this.arr.length; i++) {
9             s = s + this.arr[i];
10        }
11    }
12    . . .
13 }
```

Define Business methods

```
1 public static void main(String[] args) {  
2  
3     IntegerArray obj = new IntegerArray();  
4     obj.input("Input integer array");  
5     obj.output("Output integer array");  
6     int s = obj.sum();  
7     System.out.println("Sum of all items: " + s);  
8  
9 }
```

3

Overload Methods

Define overload methods

```
1 public class Fraction{
2     private int numerator;
3     private int denominator;
4     . . .
5     public Fraction sum(Fraction obj){
6         Fraction result = new Fraction();
7         Fraction.numerator = this.numerator * obj.denominator + this.denominator
8 * obj.numerator;
9         result.denominator = this.denominator * obj.denominator;
10        return result;
11    }
12    public Fraction sum(int value){
13        Fraction result = new Fraction();
14        result.numerator = this.numerator + this.denominator * value;
15        result.denominator = this.denominator;
16        return result;
17    }
18 }
```

Define overload methods

```
1 package Exercise01;
2 public class Main {
3     public static void main(String[] args) {
4
5         Fraction obj_1 = new Fraction(2, 3);
6         Fraction obj_2 = new Fraction(3, 4);
7
8         Fraction result_1 = obj_1.sum(obj_2);
9         result_1.output("Sum of obj_1 and obj_2");
10
11        Fraction result_2 = obj_1.sum(40);
12        result_2.output("Sum of obj_1 and 40");
13
14    }
15 }
```

Define overload methods

```
1 package Exercise03;
2 public class FractionArray {
3     private Fraction[] arr;
4     . . .
5     public Fraction sum() {
6         Fraction s = this.arr[0];
7         for(int i = 1; i < this.arr.length; i++) {
8             s = s.sum(this.arr[i]);
9         }
10    }
11    public Fraction sum(FractionArray obj) {
12        Fraction s_1= this.sum();
13        Fraction s_2= obj.sum();
14        Fraction result = s_1.sum(s_2);
15        return result;
16    }
17 }
```

Define overload methods

```
1 public static void main(String[] args) {  
2  
3     FractionArray arr_1 = new FractionArray();  
4     arr_1.input("Input Fraction array 1");  
5     Fraction s_1 = arr_1.sum();  
6     s_1.output("Sum of Fraction array 1");  
7  
8     FractionArray arr_2=new FractionArray();  
9     arr_2.input("Input Fraction array 2");  
10    Fraction s_2 = arr_1.sum(arr_2);  
11    s_2.output("Sum of Fraction array 1 & 2");  
12 }
```

Define overload methods

```
1 package Exercise02;
2 public class IntegerArray {
3     private int[] arr;
4     . . .
5     public int sum() {
6         int s = 0;
7         for(int i = 0; i < this.arr.length; i++){
8             s = s + this.arr[i];
9         }
10    }
11    public int sum(IntegerArray arr) {
12        int s_1 = this.sum();
13        int s_2 = arr.sum();
14        int result = s_1 + s_2;
15        return result;
16    }
17 }
```

Define overload methods

```
1 public static void main(String[] args) {
2
3     IntegerArray arr_1 = new IntegerArray();
4     arr_1.input("Input integer array 1");
5     int s_1 = arr_1.sum();
6     System.out.println("Sum of integer array 1 is " + s_1);
7
8     IntegerArray arr_2 = new IntegerArray();
9     arr_2.input("Input integer array 2");
10    int s_2 = arr_1.sum(arr_2);
11    System.out.println("Sum of integer array 1&2 is " + s_2);
12 }
```

4

Static Class/Methods

Define static methods

```
1 public class Fraction {  
2     private int numerator;  
3     private int denominator;  
4     . . .  
5     public static Fraction parseFraction(String str) {  
6         Fraction result = new Fraction();  
7         String [] s = str.split("/");  
8         result.numerator = Integer.parseInt(s[0]);  
9         result.denominator = Integer.parseInt(s[1]);  
10        return result;  
11    }  
12 }  
13 }
```

Define static methods

```
1 package Exercise01;
2 public class Main {
3     public static void main(String[] args) {
4
5         Fraction obj_1 = Fraction.parseFraction("2/3");
6         Fraction obj_2 = Fraction.parseFraction("3/4");
7
8         Fraction result_1= obj_1.sum(obj_2);
9         result_1.output("Sum of Fraction obj_1 and obj_2");
10
11        Fraction result_2 = obj_1.sum(40);
12        result_2.output("Sum of Fraction obj_1 and 40");
13
14    }
15 }
```

Define static methods

```
1  package Exercise03;
2  public class FractionArray {
3      private Fraction[] arr;
4      . . .
5      public static FractionArray parseFractionArray(String str) {
6          FractionArray obj = new FractionArray();
7          String[] s = str.split(";");
8          obj.arr = new Fraction[s.length];
9          for(int i = 0; i < s.length; i++){
10             obj.arr[i] = Fraction.parseFraction(s[i].trim());
11         }
12         return obj;
13     }
14 }
```

Define static methods

```
1  public static void main(String[] args) {
2
3      FractionArray obj_1 = FractionArray.parseFractionArray("1/2;2/3;3/4");
4      Fraction s_1 = obj_1.sum();
5      s1.xuat("Sum of Fraction obj_1");
6
7      FractionArray obj_2 = FractionArray.parseFractionArray("4/3;7/6;9/2");
8      obj_2.input("Input Fraction array");
9      Fraction s_2 = obj_1.sum(obj_2);
10     s_2.output("Sum of Fraction array 1 & 2");
11 }
```

Define static methods

```
1 package Exercise03;
2 public class IntegerArray {
3     private int[] arr;
4     . . .
5     public static IntegerArray parseIntegerArray(String str) {
6         IntegerArray obj = new IntegerArray();
7         String[] s = str.split(";");
8         obj.arr = new int[s.length];
9         for(int i = 0; i < s.length; i++) {
10             obj.arr[i] = Integer.parseInt(s[i].trim());
11         }
12         return msn;
13     }
14 }
```

Define static methods

```
1 public static void main(String[] args) {
2
3     IntegerArray obj_1 = IntegerArray.parseIntegerArray("2;3;4");
4     int s_1 = obj_1.sum();
5     System.out.println("Sum of integer array obj_1: " + s_1);
6
7     IntegerArray obj_2 = IntegerArray.parseIntegerArray("4;7;9");
8     int s_2 = obj_1.sum(obj_2);
9     System.out.println("Sum of integer array 1&2: " + s_2);
10 }
```



References

References

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